

CAP-01 / DOCUMENT 03

Requirements Gathering and Traceability

Shopping Copilot

Arabic-first shopping advice and guarded Storefront actions

Programme	Digital Egypt Pioneers Initiative (DEPI)
Track	AWS ML Engineering
Submission deadline	16 October 2026
Export date	6 October 2026
Source document date	2 October 2026
Release status	Review-ready export; official submission pending

Prepared from the editable project documents. Lecturer approval evidence, official organization-repository access and grading details remain pending. Nominal responsibilities are not evidence of individual contributions.

Requirements gathering and traceability

Project: Shopping Copilot. **Updated:** 2 October 2026. **Track:** AWS ML Engineering. **DEPI deadline:** 16 October 2026. **Status:** internal review draft aligned to the [supplied content guidelines](#); lecturer assessment and official upload pending.

Actual working arrangement (2 October 2026): the owner and coding assistant perform the work; the owner coordinates this deliverable. Earlier named academic leads were administrative allocations, not verified contributions or available staffing. See the updated [CAP-01 register](#).

Problem and stakeholders

A Shopper may know what they want but struggle to choose among products or translate a request into filters, navigation and cart controls. The project addresses both decision assistance and execution on one Controlled Storefront. It must support natural multilingual input while preserving the Shopper's control over consequential actions.

Stakeholders are the Shopper, the project team maintaining the catalogue/application, and the DEPI reviewers assessing the graduation deliverables. Team technical roles and confirmed dates are recorded in the [CAP-01 register](#). The deployment is a controlled graduation demonstration with fictional commerce data, not a service connected to real retailers or payment providers.

Elicitation sources and their limits

Source: Owner-approved local specification and domain model

What it establishes: Supported journeys, controlled scope and safety invariants

Limitation: Earlier wording is superseded where later owner decisions are explicit.

Source: Informal feedback from five testers, relayed by the owner

What it establishes: Positive reactions to advice; two successful account navigations; demand for natural styling and comparisons

Limitation: No standardized scores, complete task matrix or measured comparative benefit.

Source: Owner observations and approved changes

What it establishes: Need for exact-match explanations, conversational follow-up and model-owned interpretation

Limitation: Individual reports are not population-level research.

Source: Runtime tests and live reports

What it establishes: Concrete observed successes, failure modes and guarded behaviour

Limitation: Coverage is finite and tied to recorded configurations.

Source: Owner-supplied DEPI schedule and team table

What it establishes: Applicable dates and assigned technical roles

Limitation: The images do not establish official rubric weights or a document template.

Source: Five supplied DEPI PDFs, reviewed 2 October

What it establishes: Required documentation content, infographic, organization GitHub delivery and confirmed dates.

Limitation: Literature/lecturer heading discrepancy; grading weights and actual repository access not supplied.

All five testers reportedly lacked confidence choosing purchases online; this is not evidence that all had difficulty operating website controls. The recorded findings were robotic responses and insufficient comparison/decision support. UI comments were not supplied in detail and are not invented as requirements. The approved informal study and technical release exceptions are fully recorded in [Ticket 16 closeout](#).

Functional requirements

User stories and use-case mapping

These stories express requirements rather than quotations from participants. Detailed preconditions, alternative flows and observable outcomes are in the [design use cases](#).

Story: US-01

Shopper goal: As a Shopper, I want to express budget, size and exclusions in my own language so I can find suitable products.

Requirement / use case: FR-01-02 / UC-01

Acceptance example: Preserve explicit exclusions; unknown facts cannot qualify a product as an Exact Match.

Story: US-02

Shopper goal: As a Shopper, I want to discuss trade-offs so I can make a more informed choice.

Requirement / use case: FR-03 / UC-02

Acceptance example: Compare verified facts, label styling opinions, acknowledge unknown qualities; no mutation.

Story: US-03

Shopper goal: As a Shopper, I want to open an identified product or find a control so I can continue shopping.

Requirement / use case: FR-04 / UC-03

Acceptance example: Navigate only to supported observed destinations; Spotlight does not submit checkout.

Story: US-04

Shopper goal: As a Shopper, I want to edit one cart line and undo a mistake so other items stay unchanged.

Requirement / use case: FR-05 / UC-04-05

Acceptance example: Resolve the exact variant; distinguish added quantity from total; preserve the original Undo expiry.

Story: US-05

Shopper goal: As a Shopper, I want to approve the precise effect before clearing the cart or submitting fictional checkout.

Requirement / use case: FR-06 / UC-06

Acceptance example: Changed cart state or expired/reused Confirmation cannot authorize the effect.

Story: US-06

Shopper goal: As a Shopper, I want account/order guidance while entering private details myself.

Requirement / use case: FR-07 / UC-07

Acceptance example: Sensitive values are excluded from observations and model requests.

Story: US-07

Shopper goal: As a Shopper, I want to stop or refresh without accidentally repeating a cart operation.

Requirement / use case: FR-08 / UC-08

Acceptance example: Reject stale results and reconcile uncertain outcomes before any further effect.

Story: US-08

Shopper goal: As a Shopper, I want to review dictated text before sending it and type when speech is unavailable.

Requirement / use case: FR-09 / UC-09

Acceptance example: Transcription is editable; recording/availability failures do not block typing.

Functional acceptance

Priority **Must** means required within the approved graduation scope; it does not imply every future cloud or ML requirement is already implemented.

ID: FR-01

Requirement: Interpret Egyptian Arabic, Franco-Arabic, English and mixed-language shopping requests, including corrections and references.

Observable acceptance and evidence: Evaluate labelled intent/Constraints separately from completed browser tasks; report failures by language/scenario.

State: Local implementation/evidence exists; new dataset evaluation planned.

ID: FR-02

Requirement: Discover available products using explicit requirements, budget, size, colour and ordering preferences.

Observable acceptance and evidence: Verify catalogue eligibility; distinguish Exact Matches from Alternatives with named unmet or unknown requirements.

State: Implemented and tested locally.

ID: FR-03

Requirement: Provide natural advice, relevant comparisons and concise reasons for exact matches using verified catalogue facts.

Observable acceptance and evidence: Opinions are distinguishable from facts; absent qualities remain unknown; ask a useful question when needed without forcing a questionnaire.

State: Implemented; qualitative evidence and known grounding limitations recorded.

ID: FR-04

Requirement: Navigate or Spotlight the requested Storefront destination and open an identified product.

Observable acceptance and evidence: Use current observed targets and same-origin destinations; do not confuse locating a control with activating it.

State: Implemented and tested locally.

ID: FR-05

Requirement: Add a selected available variant, change quantity, remove a line and offer ten-second Undo.

Observable acceptance and evidence: Assert exact variant/quantity and unaffected lines; Undo restores the precise prior state while its original deadline remains valid.

State: Implemented; automated and recorded live checks.

ID: FR-06

Requirement: Require fresh, bound, single-use Confirmation for bulk clearing and checkout submission.

Observable acceptance and evidence: Unconfirmed, expired, mismatched or reused authorization cannot mutate state.

State: Implemented; current regressions and historical live evidence.

ID: FR-07

Requirement: Support account/order guidance and newest-order Spotlight without handling sensitive login/payment fields.

Observable acceptance and evidence: Shopper enters sensitive values directly; agent acts only on permitted controls.

State: Implemented; two reported account navigations are a narrower human observation.

ID: FR-08

Requirement: Support Stop, refresh reconciliation and task ownership across tabs.

Observable acceptance and evidence: Discard late/stale Actions and do not replay an uncertain mutation automatically.

State: Implemented and tested locally.

ID: FR-09

Requirement: Offer readable chat, RTL/localization, product cards and editable speech input when the selected service is available.

Observable acceptance and evidence: The Shopper can type when speech is unavailable and can review transcription before sending.

State: Implemented; paid voice currently constrained by provider-key allowance.

All functional requirements above are Must within supported local capabilities. They do not extend the catalogue to laptops merely because a participant used laptops as a comparison example.

Safety and quality requirements

ID: SQ-01

Requirement: The LLM owns interpretation; validation must not become a shopper-language keyword/regex interpreter.

Verification approach: Inspect boundaries and test schema/target/quantity failures independently from live language-quality cases.

ID: SQ-02

Requirement: Reject off-origin navigation and exclude Sensitive Field values from model context.

Verification approach: Adversarial browser and deterministic policy regressions.

ID: SQ-03

Requirement: Validate inventory, quantity/price bounds, observed targets and exact mutation authorization.

Verification approach: Public API, Bridge and Storefront regression checks, including stale and duplicate attempts.

ID: SQ-04

Requirement: Preserve truthful evidence and safe fallback.

Verification approach: Retain failures, unknown usage and original artifacts; malformed advice preserves verified cards without retry loops or mutation.

ID: SQ-05

Requirement: Measure latency, task steps and cost with explicit denominators.

Verification approach: Separate model-inclusive time from browser execution; count repairs and do not substitute estimates for missing billing.

ID: SQ-06

Requirement: Prevent credential and participant-data disclosure.

Verification approach: Keep environment files/secrets out of Git, omit personal contacts from public reports, and obtain authorization for recording/data reuse.

ID: SQ-07

Requirement: Keep advice read-only and resistant to untrusted instructions in product text.

Verification approach: No executable fields/tools in advice responses; later action requests re-enter normal execution guards.

Current evidence is summarized in [RESULTS.md](#). The historical 44-case full-model gate and its performance definitions remain tied to its frozen runtime. The local release uses the owner's explicit evidence exception; it does not prove every paraphrase, product claim or browser interaction is correct.

Graduation ML and deployment requirements

ID: GR-01

Requirement: Release reviewed multilingual data with provenance and exposure history.

Planned acceptance: CAP-02: documented labels, reviewer decisions, group-aware splits and an untouched evaluation set.

ID: GR-02

Requirement: Train an offline classical baseline and compare it fairly with the LLM.

Planned acceptance: CAP-03: TF-IDF + Logistic Regression, train-only learned transforms, validation-based selection, same eligible inputs, confusion matrices and error/cost/latency analysis.

ID: GR-03

Requirement: Deploy the Controlled Storefront application on AWS.

Planned acceptance: CAP-04-07: reviewed budget/design, qualified model, protected deployment, operational acceptance and repeatable recovery/rollback evidence.

ID: GR-04

Requirement: Qualify the AWS model before relying on it for actions.

Planned acceptance: The approved gate retains 100% schema/currency/safety compliance and at least 95% exact intent-and-Constraint accuracy on its defined corpus; report unseen results separately.

ID: GR-05

Requirement: Provide traceable academic and individual evidence.

Planned acceptance: Confirmed owners, deadlines, weekly contribution records and final documentation/demo aligned to the actual DEPI requirements.

These are Must graduation requirements and remain planned beyond the completed local MVP. The offline baseline does not enter the runtime or authorize Actions. Separate evaluation of context-dependent requests avoids claiming that isolated text classification measures full conversational execution.

Exclusions, assumptions and open questions

Excluded: arbitrary retailers, external store integrations, real transactions, multi-tenant SaaS and general production-readiness claims. AWS access, region/model eligibility and budget need confirmation before provisioning. Test fixtures and local accounts remain fictional; the current in-memory state is not assumed durable after cloud deployment.

The owner coordinates the overall submission. The supplied instructions identify organization-owned GitHub delivery, but the invitation/URL and editable report template remain unprovided. Actual work is performed by the owner and coding assistant; nominal academic roles do not establish independent review. Dataset review/custody arrangements and numerical rubric weights remain unresolved. These gaps do not change the confirmed DEPI deadlines. Requirement changes should record their source, effect on implementation/evidence and explicit approval; passed tests do not silently redefine a requirement.

Export references and provenance

Source content dates are preserved. This PDF was exported on 6 October 2026; it is not an approval or official-upload record. Repository links below use the fixed development snapshot 85369325942010639ab6a765d8402e443beeff1a. They identify supporting evidence, not the official submission destination. External bibliography URLs are preserved as supplied and were not re-verified during export.

Editable source: docs/cap01-requirements.md

SHA-256: 9e16a19d6b26896cf007f3b66ba593ed53575c65d23c28404a0334cfc9c8555a

Linked sources

1. supplied content guidelines

<https://github.com/Homd11/shopping-copilot/blob/85369325942010639ab6a765d8402e443beeff1a/docs/depi-guideline-alignment.md>

2. CAP-01 register

<https://github.com/Homd11/shopping-copilot/blob/85369325942010639ab6a765d8402e443beeff1a/docs/cap01-execution.md>

3. Ticket 16 closeout

<https://github.com/Homd11/shopping-copilot/blob/85369325942010639ab6a765d8402e443beeff1a/docs/ticket16-closeout.md>

4. design use cases

<https://github.com/Homd11/shopping-copilot/blob/85369325942010639ab6a765d8402e443beeff1a/docs/system-analysis-design.md#3-use-cases>

5. RESULTS.md

<https://github.com/Homd11/shopping-copilot/blob/85369325942010639ab6a765d8402e443beeff1a/RESULTS.md>